

Q1

Considering a computer system with a 10-bit logical address. Translate the logical address 1000011011 to the corresponding physical address for the following two cases:

- a) Assuming paging is used, the page size is 128 bytes and the page table is as given in Figure Q4a.
- b) Assuming segmentation is used, the maximum segment size that the system can support is 256 bytes, and the segment table is as given in Figure Q4b.

0	01011
1	00010
2	00010
3	10101
4	01001
5	01100
6	11010
7	00110

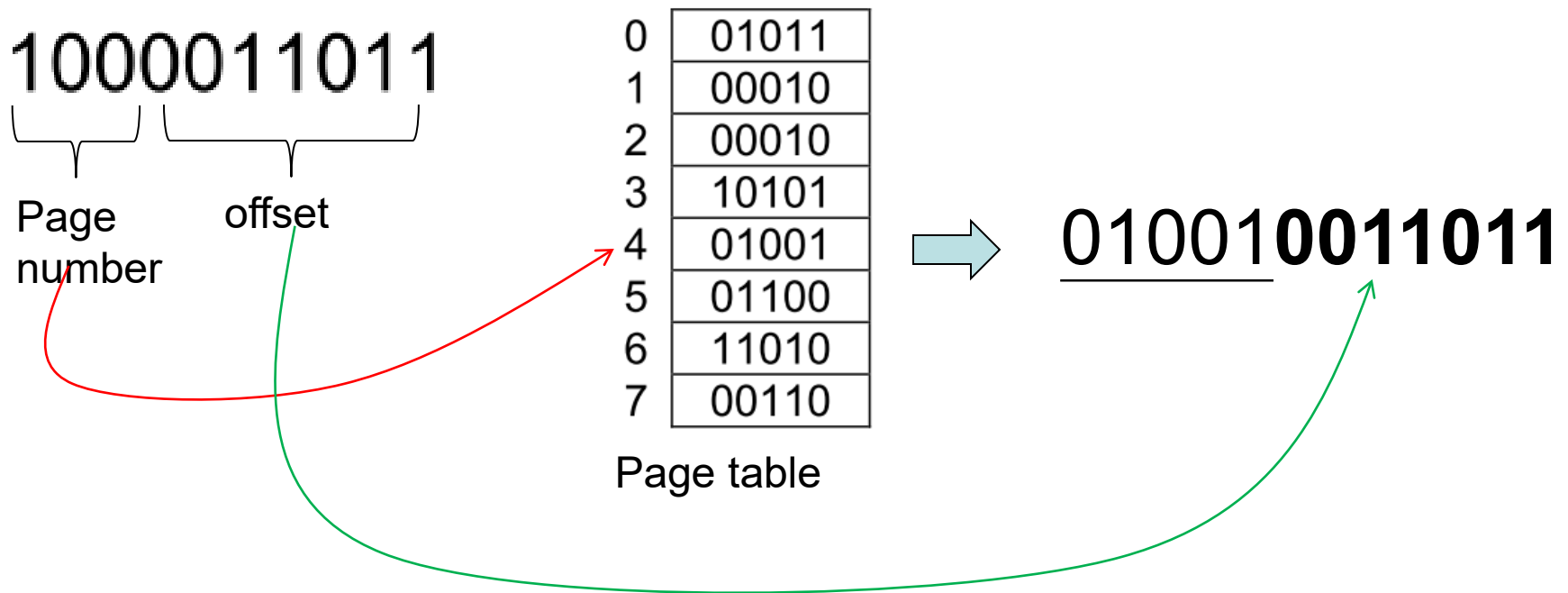
Figure Q4a

0	00010000	010000000000
1	00001000	100000000000
2	00100000	010011010000
3	00011000	100001100000

Figure Q4b

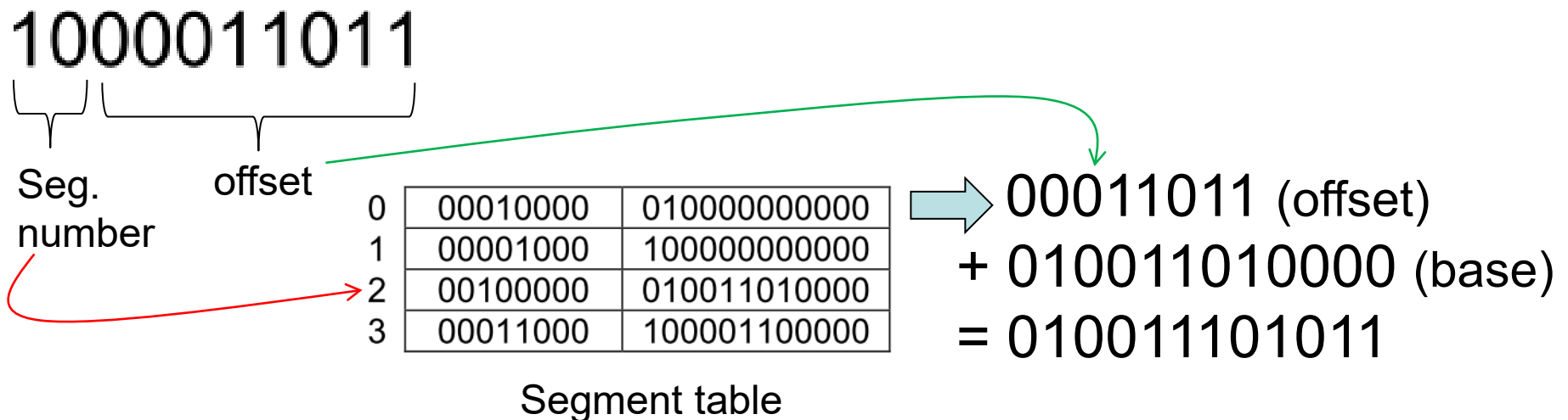
Q1 (a)

- Given page size is 128 bytes → 7 bits are needed for page offset and 3 bits for page number.



Q1 (b)

- The maximum segment size is 256 bytes → 8 bits are needed for the segment offset and 2 bits for segment number.



Q2

Assume a computer system uses paging for memory allocation. Logical address consists of x bits for the page number and y bits for the offset.

- a) What is the page size?
- a) How many entries are there in the page table?
- a) Assuming each page table entry occupies 4 bytes, how many entries are there in the outer level page table if a two-level paging scheme is used? What is the format of logical address?

Q2 (Answer)

- a) Page size: 2^y bytes
- b) Number of entries in page table: 2^x entries
- c) Number of entries in the outer level page table: $(2^x \times 4)/2^y = 2^{x-y} \times 4$

Number of entries per page in a page table: $2^y/4$

Logical address format: $x-y+2 \mid y-2 \mid y$

Q3

For each of the page replacement policies listed below, calculate the number of page faults encountered when referencing the following pages. Assume the availability of 4 empty page frames.

0 1 6 0 3 4 0 1 0 3 4 6 3 4

a) FIFO

b) CLOCK

c) LRU

Q3 (FIFO)

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	4	4	4	4	4	4	4	3	3
F2		1	1	1	1	1	0	0	0	0	0	0	0	4
F3			6	6	6	6	6	1	1	1	1	1	1	1
F4					3	3	3	3	3	3	3	6	6	6
						0	1	6				3	4	0
	P	P	P		P	P	P	P				P	P	P

There are 10 page faults.

Q3 (CLOCK)

Page Number R Bit Circular Queue

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0,0	0,0	0,0	0,1	0,1	0,0	0,1	0,1	0,1	0,1	0,1	0,0	0,0	0,0
F2		1,0	1,0	1,0	1,0	4,0	4,0	4,0	4,0	4,0	4,1	4,0	4,0	4,1
F3			6,0	6,0	6,0	6,0	6,0	1,0	1,0	1,0	1,0	6,0	6,0	6,0
F4					3,0	3,0	3,0	3,0	3,0	3,1	3,1	3,0	3,1	3,1
						1		6				1		
	P	P	P		P	P		P				P		

There are 7 pages faults.

Clock hand in shaded entry.

Q3 (LRU)

0 1 6 0 3 4 0 1 0 3 4 6 3 4

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F2		1	1	1	1	4	4	4	4	4	4	4	4	4
F3			6	6	6	6	6	1	1	1	1	6	6	6
F4					3	3	3	3	3	3	3	3	3	3
						1		6				1		
	P	P	P		P	P		P				P		

There are 7 page faults.